
A Shared Invertible Latent for Cross-Modal Medical Image Synthesis

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Abstract

1 Multi-modal medical imaging captures complementary views of the same pathol-
2 ogy, motivating representations that are simultaneously *modality-agnostic*, *genera-*
3 *tive*, and *directly usable* for downstream analysis. Replacing per-modality encoders
4 and decoders with a single normalizing flow attains an exact shared latent but
5 discards the convolutional inductive biases that medical images rely on, whereas
6 training independent models per modality yields incompatible latent geometries.
7 We present a hybrid that keeps a per-modality convolutional VAE and bridges the
8 modality-specific latents through per-modality normalizing flows into one shared
9 low-dimensional latent $\mathbf{u}$, on which a Gaussian-mixture prior is placed. Because the
10 flows are invertible, cross-modal translation is exact and routed entirely through $\mathbf{u}$;
11 because $\mathbf{u}$ is a native encoder output rather than a probed intermediate activation, it
12 doubles as a representation for clustering and cross-modal alignment. On real ADNI
13 T1 $\leftrightarrow$ FA (diffusion-derived fractional anisotropy), our model surpasses faithfully
14 reproduced CFM, DDPM, DiT, and CycleGAN baselines on cross-modal SSIM
15 and PSNR in both directions. Our central contribution is a controlled analysis of
16 *why* such a shared latent aligns: through a reconstruction-collapse ladder we show
17 that cross-modal alignment is conferred neither by the mixture prior, the KL term,
18 nor the flow likelihood, but by two necessary and jointly sufficient ingredients—a
19 reconstruction-dominant objective and a low-dimensional bottleneck—isolating a
20 failure mode common to multi-term latent-alignment models. We further find that
21 the learned clusters are geometrically real and modality-consistent yet not aligned
22 with diagnosis, a limit we trace to the evaluation cohort rather than the model,
23 and we quantify the perception–distortion trade-off underlying the reconstruction
24 metrics. Together these results give a compact, invertible recipe for cross-modal
25 synthesis and a clear account of what does and does not drive latent alignment.

26 1 Results

27 2 Ablations

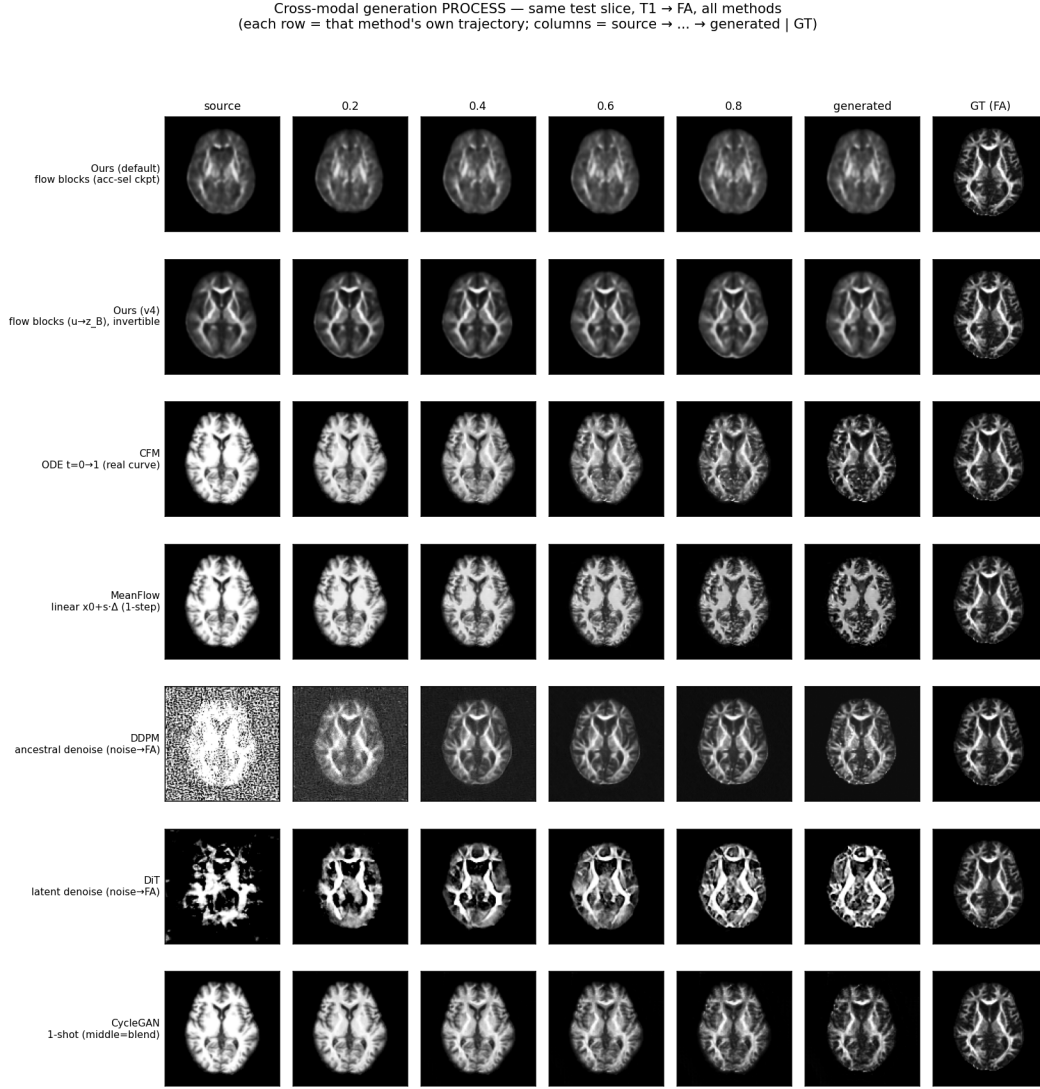


Figure 1: Cross-modal generation process (T1 $\rightarrow$ FA), all methods; rows are each method’s own trajectory, source to generated with GT at right. Our invertible shared-latent flow (top rows) transfers through a single $\mathbf{u}$.

Table 1: Cross-modal synthesis (SSIM/PSNR).

Method	T1 $\rightarrow$ FA		FA $\rightarrow$ T1	
	SSIM	PSNR	SSIM	PSNR
CycleGAN	0.680	17.18	0.743	17.94
DDPM	0.362	18.44	0.769	20.36
DiT	0.665	14.32	0.757	18.28
MeanFlow	0.686	17.37	0.621	17.27
CFM	0.732	18.46	0.802	19.99
Ours	0.764	19.95	0.859	22.05

Table 2: Clustering ($K=4$). Sil $\uparrow$, DB $\downarrow$.

Method	ACC	NMI	subj-CCA	Sil	DB
Majority baseline	0.395	—	—	—	—
naive VAE	0.442	0.112	0.047	—	—
CycleGAN	0.395	0.105	—	—	—
DDPM	0.395	0.088	—	—	—
MeanFlow	0.372	0.098	—	—	—
DiT	0.442	0.091	—	—	—
CFM	0.419	0.098	—	—	—
Ours (default)	0.442	0.099	0.023	0.130	2.207
Ours (recon-opt)	0.349	0.059	0.628	0.181	1.760

Table 3: Unified ablation. All variants on shared axes: bottleneck dim, reconstruction regime (mean vs. sum-dominant), self-recon SSIM (a collapse indicator: 0.68=mean-brain), cross-modal SSIM, alignment CCA (chance 0.25), and GMM cluster quality (silhouette Sil, Davies–Bouldin DB). All recon numbers are SSIM (PSNR in Table 1). Alignment appears *only* with sum-recon *and* low dim; removing cross-recon breaks it; GMM strength moves Sil/DB, not synthesis.

Variant	dim	recon	SSIM			align CCA	cluster	
			self	T1→FA	FA→T1		Sil↑	DB↓
<i>Collapse ladder (add one ingredient; balanced weights)</i>								
bare-align	100352	mean	0.68	—	—	—	—	—
+cross/flow/GMM/KL	≤100352	mean	0.68	—	—	0.00	—	—
flow-NLL ×10/ ×1000	16	mean	0.68	—	—	0.00	—	—
A1/B1 (recon-dominant, hi-dim)	100352	sum	0.89	—	—	0.00	—	—
<i>Loss deletion (sum-recon, dim 8)</i>								
full	8	sum	0.89	0.768	0.862	0.938	—	—
—GMM-NLL	8	sum	0.89	0.769	0.862	0.952	—	—
—flow-NLL	8	sum	0.89	0.769	0.862	0.948	—	—
—KL	8	sum	0.90	0.768	0.863	0.948	—	—
—paired-align	8	sum	0.89	0.767	0.863	0.951	—	—
— cross-recon	8	sum	0.90	0.724	0.797	0.802	—	—
<i>GMM flow-base, 3-stage (sum-recon; Sil(dx)< 0 for all)</i>								
v2 (weak GMM)	8	sum	0.89	0.770	0.863	0.94	0.314	1.211
v4 (strong S1-2, weak S3)	8	sum	0.89	0.768	0.863	0.94	0.305	1.137
v5 (16d, from scratch)	16	sum	0.90	0.765	0.864	0.94	0.264	1.371
v1 (strong GMM)	8	sum	0.88	0.763	0.854	0.93	0.585	0.579

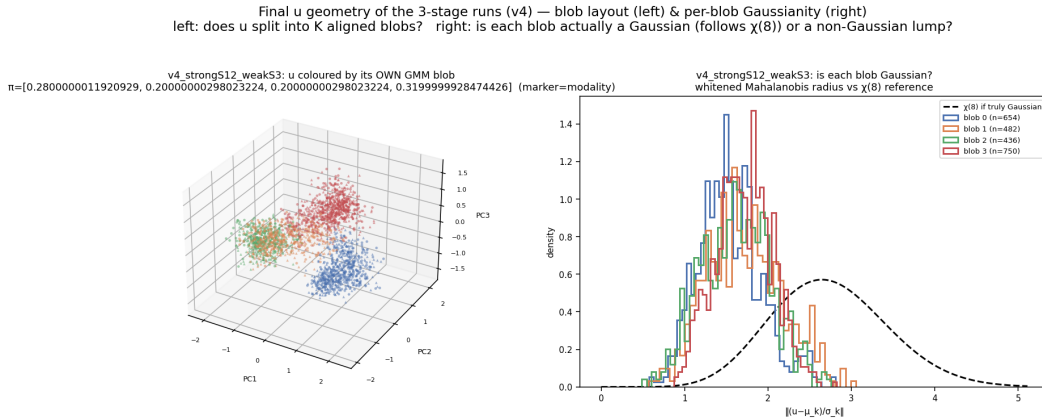


Figure 2: Shared latent u (our run): 3-D layout coloured by GMM cluster (left) and per-cluster Gaussianity vs. $\chi(8)$ (right). Clusters are real but sub-Gaussian.

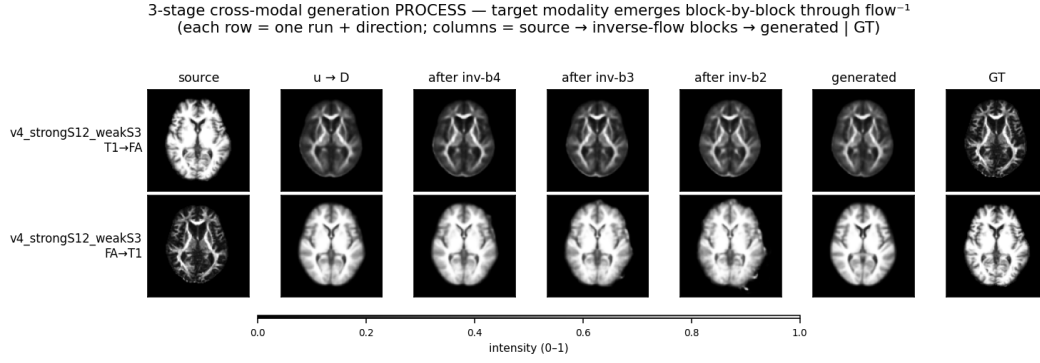


Figure 3: Cross-modal generation emerges block-by-block through flow^{-1} (our run, both directions), staying on-manifold and progressive.

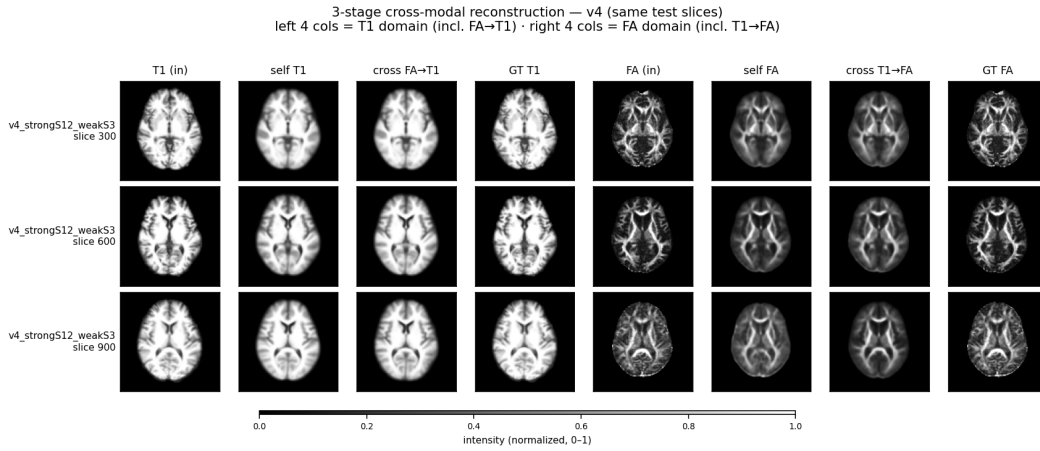


Figure 4: Bidirectional reconstruction (our run, 3 test slices, shared colourbar). Self-recon is bottleneck-limited; cross-modal detail is bounded by what the source modality carries.

Step-by-step cross-modal transfer through normalizing flow — v4_strongS12_weakS3, by disease case
 (left: T1 morphing as flow_A pushes $z \rightarrow u$ | right: FA emerging as flow_B⁻¹ pulls $u \rightarrow z_B$)

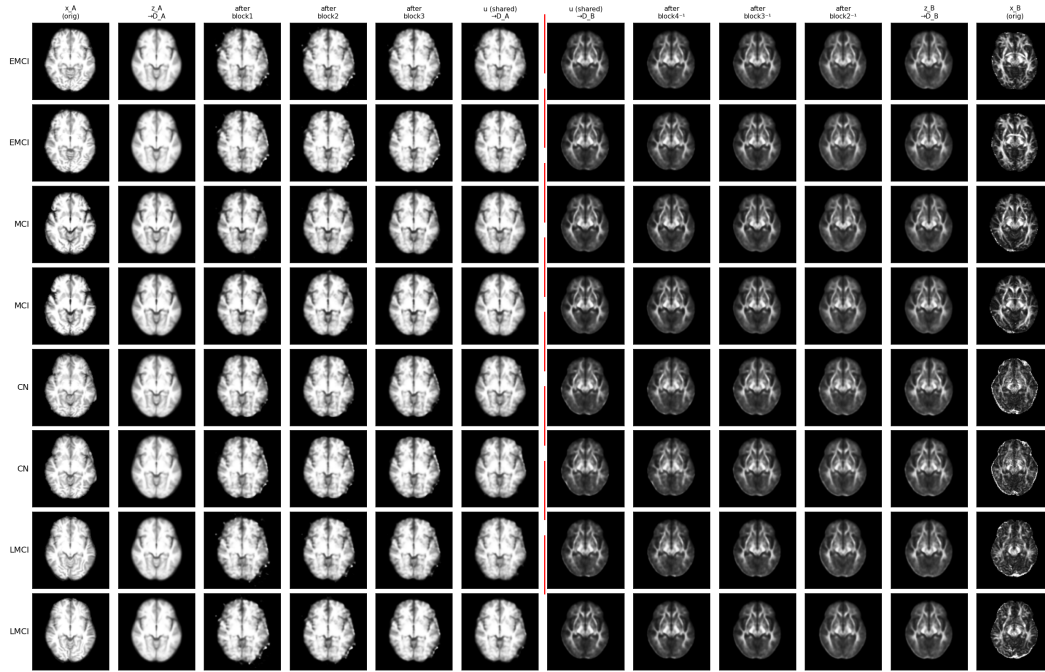


Figure 5: Full flow trace by disease case (run v4). Left: T1 morphs as flow_A pushes $z \rightarrow u$; right: FA emerges as flow_B⁻¹ pulls $u \rightarrow z_B$.